

Recall:

Linear binary code with length  $n$  and dimension  $k$   
 $\Leftrightarrow k$ -dimensional subspace of  $\mathbb{F}_2^n$

Basis for a linear code  $C$  is a subset  $B \subseteq C$  such that

- $\langle B \rangle = C$
- $B$  is a linearly independent set

A generating matrix for a linear code  $C$  of length  $n$  and dimension  $k$  is a  $k \times n$  matrix whose rows form a basis for  $C$ .

There are many different generating matrices for a code  $C$ , but only one in RREF.

A linear code generated by a generating matrix of the form  $G = (I_k | X)$  is called systematic.

Messages  $m \in \mathbb{F}_2^k$  are encoded as  $mG$ .

Exercise Consider the linear code  $C = \langle S \rangle$  for  
 $S = \{1100111, 1101010, 1011001, 0111110\}$ .

(a) Determine the generating matrix in rref for  $C$ .

(b) What is the length  $n$  and dimension  $k$  for  $C$ .

(c) Is the code  $C$  systematic?

(d) Assuming that your matrix from part (a) was used for the encoding process, determine the message word corresponding to codeword 0111110.

$$\begin{pmatrix} 1 & 1 & 0 & 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 1 & 1 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & 1 & 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$$G = \begin{pmatrix} 1 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 & 1 \end{pmatrix}$$

$n=7$   
 $k=3$   
 not systematic

$mG = 0111110$   
 $m = 011$

$G' = (I | X) = \begin{pmatrix} 1 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 \end{pmatrix}$   
 swap 3 & 4

## 4.6. Dual Codes

We shall now see how to derive a new code from a given linear code  $C$ , using the orthogonal complement. A generating matrix for this new code is the matrix  $H$ , which is very useful for error detection and correction in the original code  $C$ .

Recall that vectors  $v, w \in \mathbb{F}_2^n$  are *orthogonal* if  $v \cdot w = 0 \in \mathbb{F}_2$ . We write  $v \perp w$ .

If  $\emptyset \neq S \subseteq \mathbb{F}_2^n$ , the set of vectors orthogonal to every vector in  $S$  is denoted  $S^\perp$ .

$$S^\perp = \{v \in \mathbb{F}_2^n \mid v \cdot s = 0 \text{ for all } s \in S\}.$$

Exercise Let  $S = \{110, 011\}$ . Determine  $S^\perp$ .

$$\begin{aligned} (a \ b \ c) \cdot (1 \ 1 \ 0) &= 0 & a &= b \\ (a \ b \ c) \cdot (0 \ 1 \ 1) &= 0 & b &= c \end{aligned}$$

$$S^\perp = \{000, 111\}$$

**4.6.2 Lemma** For any non-empty set  $S \subseteq \mathbb{F}_2^n$ , the set  $S^\perp$  is a vector space (linear code).

$$\begin{aligned} &\text{if } u, v \in S^\perp \text{ and } \text{let } s \in S & 0 \in S^\perp \\ &(u+v) \cdot s = u \cdot s + v \cdot s = 0 + 0 = 0 \end{aligned}$$

**4.6.3 Lemma** If  $S \subseteq \mathbb{F}_2^n$ , then  $\langle S \rangle^\perp = S^\perp$ . *Proof in notes*

$$\rightarrow \text{eg. } \langle S \rangle = \{110, 011, 101, 000\}$$

**4.6.4 Lemma** Suppose  $U, V$  are subspaces of  $\mathbb{F}_2^n$ ,  $S$  is a basis for  $U$ ,  $T$  is a basis for  $V$  and  $s \cdot t = 0$  for all  $s \in S$  and  $t \in T$ . Then  $u \cdot v = 0$  for all  $u \in U$  and all  $v \in V$ .

*Proof* Let  $u \in U$  and  $v \in V$ . Then  $u = s_1 + \cdots + s_p$  and  $v = t_1 + \cdots + t_q$  where  $s_1, \dots, s_p \in S$  and  $t_1, \dots, t_q \in T$ . Thus,  $u \cdot v = (s_1 + \cdots + s_p) \cdot (t_1 + \cdots + t_q)$ . Since each term in the expansion of  $(s_1 + \cdots + s_p) \cdot (t_1 + \cdots + t_q)$  is 0 we have  $u \cdot v = 0$ .  $\square$

**4.6.5 Definition** If  $C \subseteq \mathbb{F}_2^n$  is a code (linear or not), the set  $C^\perp$  is called the dual code of  $C$ .

Suppose  $C$  is a linear code generated by  $S \subseteq \mathbb{F}_2^n$ . That is,  $C = \langle S \rangle$ . Then by Lemma 4.6.3 we have  $C^\perp = \langle S \rangle^\perp = S^\perp$ . Thus,

$$(4.6.1) \quad C = \langle S \rangle \implies C^\perp = S^\perp.$$

**4.6.6 Theorem** Let  $C \subseteq \mathbb{F}_2^n$  be a linear code. Then  $\dim C + \dim C^\perp = n$ .

*Proof* Let  $\dim C = k$ . Let  $S = \{c_1, \dots, c_k\}$  be a basis for  $C$ . By Equation 4.6.1

$$x \in C^\perp \iff c_j \cdot x = 0 \quad \text{for } j = 1, \dots, k.$$

Put the vectors  $c_j$  in the rows of a  $k \times n$  matrix  $A$ . Since  $A$  has linearly independent rows,  $\text{rank}(A) = \dim C = k$ . Now,  $x \in C^\perp \iff Ax^T = \mathbf{0}$ . So  $\dim C^\perp = \dim \text{NS}(A)$ . Thus, from Theorem 4.1.5,  $n = \text{rank}(A) + \dim \text{NS}(A) = \dim C + \dim C^\perp$ .  $\square$

$$\begin{matrix} & A & & x^T \\ \left( \begin{array}{c} \text{--- } c_1 \text{ ---} \\ \text{--- } c_2 \text{ ---} \\ \vdots \\ \text{--- } c_k \text{ ---} \end{array} \right) & \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} & = & \mathbf{0} \end{matrix}$$

Exercise Let  $C$  be a <sup>binary</sup>  $(10, 4)$ -linear code.

How many codewords in  $C$ ?  $2^4 = 16$

How many codewords in  $C^\perp$ ?  $2^6 = 64$

**4.6.8 Lemma** If  $V \subseteq \mathbb{F}_2^n$  is a subspace then  $(V^\perp)^\perp = V$ .

Pf  $\forall x \in V, y \in V^\perp \quad y \cdot x = 0 \text{ and } x \cdot y = 0 \quad \text{so } x \in (V^\perp)^\perp$   
 $\therefore V \subseteq (V^\perp)^\perp$

$\left. \begin{array}{l} \dim V + \dim V^\perp = n \\ \dim V^\perp + \dim (V^\perp)^\perp = n \end{array} \right\} \text{ so } \dim (V^\perp)^\perp = \dim V$

$$\begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{pmatrix} \quad \begin{pmatrix} | & | & \dots & | \\ d_1 & d_2 & \dots & d_{n-k} \\ | & | & \dots & | \end{pmatrix}$$

$G \quad k \times n \quad H \quad n \times (n-k)$

**4.6.9 Lemma** Let  $C$  be a linear  $(n, k)$ -code with generating matrix  $G$ . Suppose matrix  $H$  is an  $n \times (n-k)$  matrix with linearly independent columns. Then the columns of  $H$  form a basis for  $C^\perp$  iff  $GH = 0$ .

*Proof*  $\implies$  If the columns of  $H$  form a basis for  $C^\perp$ , then each column  $h$  of  $H$  lies in  $C^\perp$ , and so  $c \cdot h = 0$  for all  $c \in C$ . But this means  $GH = 0$ .

$\Leftarrow$  Suppose  $GH = 0$ . Let  $D$  be the span of the columns of  $H$ . By assumption, the dot product of each row vector of  $G$  with each column vector of  $H$  is 0. Thus  $u \cdot v = 0$  for each  $u$  in [a basis for]  $C$  and  $v$  in [a basis for]  $D$ . By Lemma 4.6.4, every vector in  $D$  is perpendicular to every vector in  $C$ , so  $D \subseteq C^\perp$ . But  $H$  has rank  $n-k$ , so applying Theorems 4.1.3 and 4.6.6  $\dim D = n-k = n - \dim C = \dim C^\perp$ , so  $D = C^\perp$ . That is, the columns of  $H$  span  $C^\perp$ , and they are linearly independent, so form a basis.  $\square$

**4.6.10 Definition** Let  $C$  be a linear  $(n, k)$ -code. An  $n \times (n-k)$  matrix  $H$  is called a parity check matrix for  $C$  if its columns form a basis for  $C^\perp$ .

Since the same space may have many bases, the matrix  $H$  is not unique.

Equivalently, an  $n \times (n-k)$  matrix  $H$  with linearly independent columns is a parity check matrix for  $C$  if  $GH = 0$  where  $G$  is a generating matrix for  $C$ .

If  $C$  is a systematic code with generating matrix  $G$  in standard form, then it is easy to find a parity check matrix  $H$ . Let

$$G = \begin{pmatrix} I_k & X \end{pmatrix}$$

$\uparrow \quad \uparrow$   
 $k \times k \quad k \times (n-k)$

where  $X$  is some  $k \times (n-k)$  matrix. Let

$$H = \begin{pmatrix} X \\ I_{n-k} \end{pmatrix}$$

$n \times (n-k)$

Then:  $H$  is  $n \times (n-k)$ , and the presence of  $I_{n-k}$  inside  $H$  ensures that the  $n-k$  columns of  $H$  are linearly independent. Finally by block multiplication:

$$GH = \begin{pmatrix} I_k & X \end{pmatrix} \begin{pmatrix} X \\ I_{n-k} \end{pmatrix} = I_k X + X I_{n-k} = 0_{k \times (n-k)}.$$

By Lemma 4.6.9, the columns of  $H$  form a basis for  $C^\perp$ .

## Exercise

Let  $S = \{10010, 10111, 11001, 11100\}$  and let  $C = \langle S \rangle$ .

(a) Find the generating matrix in rref for  $C$ .

(b) Is  $C$  a systematic code?

(c) What are the parameters  $(n, k, \delta)$  for  $C$ ?

(d) Find a parity check matrix for  $C$ .

(e) How many codewords in  $C^\perp$ ?

$$a) \begin{pmatrix} 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \quad G = \begin{pmatrix} 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 \end{pmatrix}$$

rref

$I_3$     $X$

(b) yes

(c)  $n = 5$     $k = 3$     $\delta = 2$

$$(d) H = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \\ 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} X \\ I_2 \end{pmatrix}$$

(e)  $\dim 2$

so 4 codewords

Even if  $C$  is not systematic we can still find a parity check matrix  $H$  as follows:

- ★ Find a systematic code  $C'$  equivalent to  $C$ , via some permutation  $C \xrightarrow{\sigma} C'$ .
- ★ If  $G$  and  $G'$  are generating matrices for  $C$  and  $C'$  respectively, then  $G \xrightarrow{\sigma} G'$  by permuting columns.
- ★ Since  $G'$  is systematic, find corresponding  $H'$  as above. Thus  $H'$  is a parity check matrix for  $C'$ .
- ★ Finally apply the inverse permutation  $\sigma^{-1}$  to the rows of  $H'$  to obtain  $H$ .

**4.6.15 Example** If  $S = \{101010, 010101, 111111, 000111, 101100\}$ , find a basis for  $C = \langle S \rangle$ , and a basis for  $C^\perp$ .

**Solution:**

$$A = \begin{pmatrix} 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 0 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

$$\rightarrow \begin{pmatrix} 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

So  $G = \begin{pmatrix} 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$ . A basis for  $C$  is  $\{101010, 010010, 000110, 000001\}$ , so  $\dim C = 4$   
 and  $|C| = 2^4 = 16$ .

$$G' = \left( \begin{array}{cccc|cc} 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \end{array} \right) = (I_4 \quad X), \quad X = \begin{pmatrix} 1 & 1 \\ 1 & 0 \\ 1 & 0 \\ 0 & 0 \end{pmatrix}.$$

Permutation  $\sigma: 3 \mapsto 6, \quad 6 \mapsto 4, \quad 4 \mapsto 3$  also written  $\sigma = (3 \ 6 \ 4)$ .

So  $\sigma^{-1}: 3 \mapsto 4, \quad 4 \mapsto 6, \quad 6 \mapsto 3, \quad \sigma^{-1} = (3 \ 4 \ 6)$ , so

$$H' = \begin{pmatrix} 1 & 1 & & & & \\ 1 & 0 & & & & \\ 1 & 0 & & [3] & & \\ 0 & 0 & & [4] & & \\ \hline 1 & 0 & & & & \\ 0 & 1 & & & [6] & \end{pmatrix} \quad H = \begin{pmatrix} 1 & 1 & & & & \\ 1 & 0 & & & & \\ 0 & 1 & & [6 \mapsto 3] & & \\ 1 & 0 & & [3 \mapsto 4] & & \\ 1 & 0 & & & & \\ 0 & 0 & & & [4 \mapsto 6] & \end{pmatrix}.$$

$GH = 0$

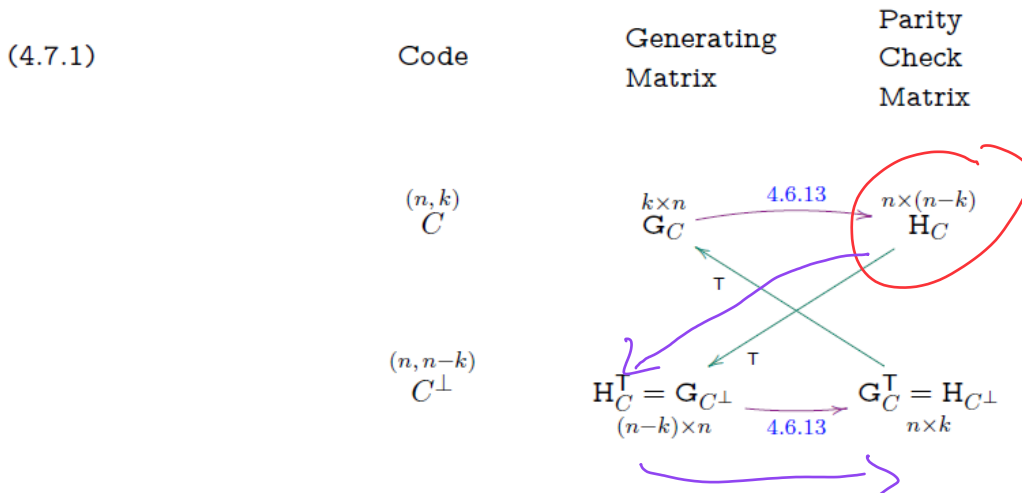
A basis for  $C^\perp$  is  $\{110110, 101000\}$ , and  $\dim C^\perp = 2$ , so  $|C^\perp| = 2^2 = 4$ .

## 4.7. Parity Check Matrices

**4.7.1 Theorem** A matrix  $H$  is a parity check matrix for a linear code  $C$  iff  $H^T$  is a generating matrix for  $C^\perp$ .

*Proof* By definition,  $H$  is a parity check matrix for  $C$  iff the columns of  $H$  are a basis for  $C^\perp$ , iff the rows of  $H^T$  are a basis for  $C^\perp$ , iff  $H^T$  is a generating matrix for  $C^\perp$ .  $\square$

So  $G_{C^\perp} = H_C^T$ . Replacing  $C$  by  $C^\perp$  and using  $(C^\perp)^\perp = C$ , we can write  $G_C = H_{C^\perp}^T$ , so  $H_{C^\perp} = G_C^T$ . This is summarized in the following important diagram:



Given any one of the generating or parity check matrices for  $C$  or  $C^\perp$ , using diagram 4.7.1 we can find the other three matrices. This means: We can specify  $C$  and/or  $C^\perp$  by giving  $H$  or  $G$ .

Exercise :  $H_C = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$  is a parity check matrix for a code  $C$ . Find a parity check matrix for the code  $C^\perp$ .

$$G_{C^\perp} = H_C^T = \begin{pmatrix} 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

$$G' = \begin{pmatrix} 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{pmatrix} = (I \ X) \quad H' = \begin{pmatrix} X \\ I \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{pmatrix} \quad H_{C^\perp} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \\ 1 & 0 \\ 0 & 0 \end{pmatrix}$$

check  $G_{C^\perp} H_{C^\perp} = 0$